

SUSTAINABILITY & INFRASTRUCTURE TRACK

Grid Failure Advisor

AI-Powered Predictive Maintenance
for Electric Grid Assets

Team · Fantastic Four · Built in 11 hours

Asset Risk Overview

All monitored grid assets, ranked by predicted failure risk.



REGION		RISK TIER					50 of 50 assets	
All		All						
NAME	TYPE	REGION	TIER	RISK SCORE ↓	GRID IMPACT	CUSTOMERS		
Eastgate Transformer 04	transformer	Eastgate	CRITICAL	100	7	14,612		
North Valley Substation 11	substation	North Valley	CRITICAL	100	8	48,708		
Eastgate Transformer 03	transformer	Eastgate	CRITICAL	97.6	7	8,453		
North Valley Transformer 04	transformer	North Valley	CRITICAL	92.6	7	6,477		
Eastgate Transformer 02	transformer	Eastgate	CRITICAL	90.5	7	4,108		
Highland Substation 03	substation	Highland	CRITICAL	84.4	8	41,031		
Bayview Transformer 05	transformer	Bayview	CRITICAL	76.1	6	13,616		
Bayview Transformer 03	transformer	Bayview	HIGH	74	6	11,032		

Grid failures are predictable.

Most utilities still act like they're not.



PILLAR 01

Calendar-Based Maintenance

Transformers are inspected every N years regardless of actual condition. Assets degrade between scheduled visits — and fail without warning.



PILLAR 02

Data Silos

Sensor data (temp, vibration, discharge, oil quality) and weather forecasts live in separate systems, never fused into one prioritized decision.



PILLAR 03

Enormous Cost of Failure

A single major transformer failure causes blackouts costing **\$1M+ per hour** and can cut power to hospitals, water plants, and thousands of homes.

The raw data to prevent this already exists. It just isn't connected.

One dashboard. Three data sources.

One ranked action list.

INPUT · 01

30-Day Sensor Data

Temp · Vibration · Discharge · Oil

INPUT · 02

7-Day Weather Forecasts

Storm Warnings · Wind · Heat

INPUT · 03

Historical Incident Records

Asset Age · Type · Past Failures

RANK · EXPLAINABLE

Explainable Risk Scoring 0-100

0-100

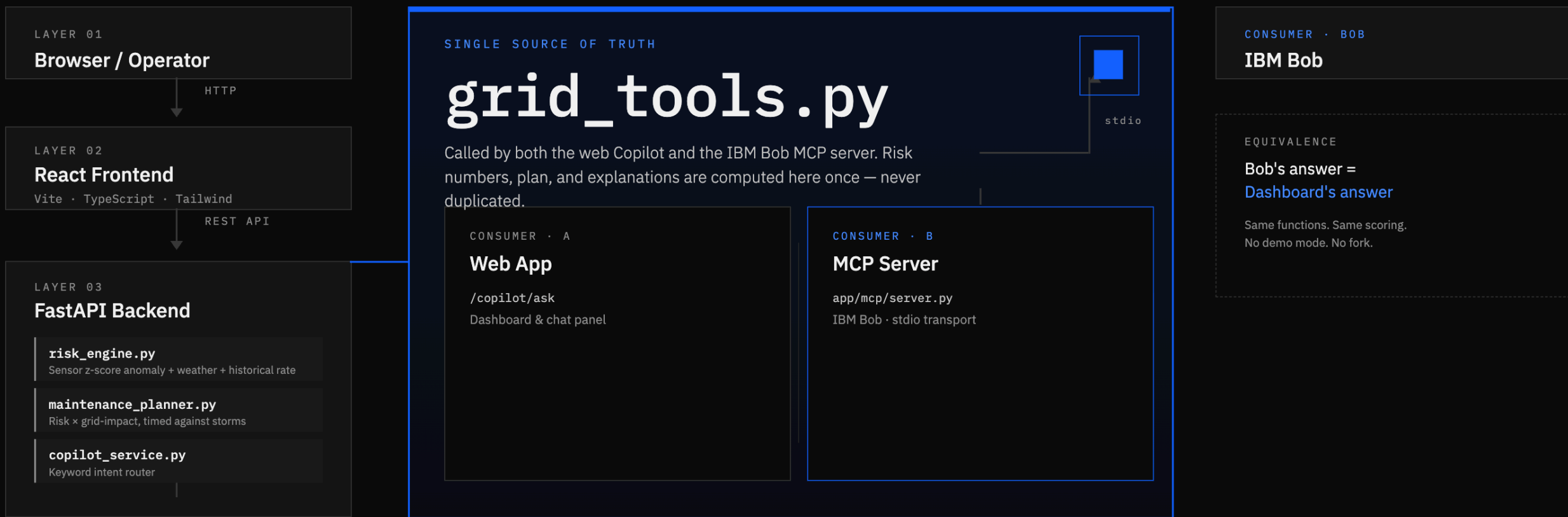
OUTPUT

Prioritized Maintenance Plan — timed before the storm hits

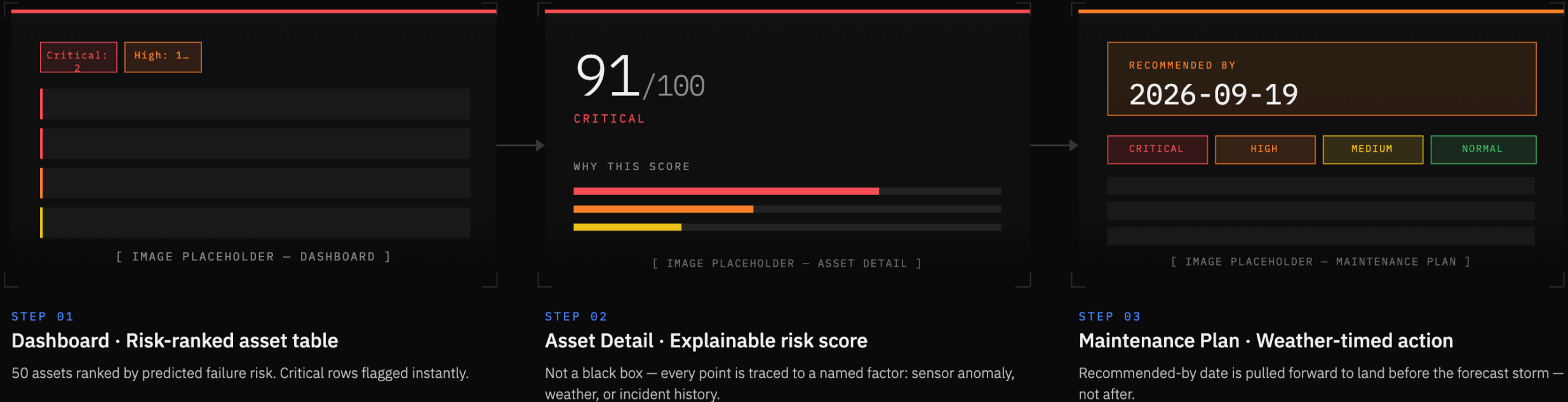
One ranked list.
Every score explained.

Clean full-stack architecture.

One shared truth for the app and IBM Bob.



From degrading sensor to dated crew dispatch — in 3 clicks.



Ask the Grid Copilot in plain English — it calls the same backend tools and shows you which ones.

[← Back to dashboard](#)

Bayview Transformer 03

transformer

Bayview

2016

22.8 MVA

11,032

6/10

74/100

- HIGH

Why This Score

Sensor Anomaly – Partial Discharge

+18 pts

Sensor Anomaly – Oil Quality

+18 pts

Sensor Anomaly – Vibration

+10 pts

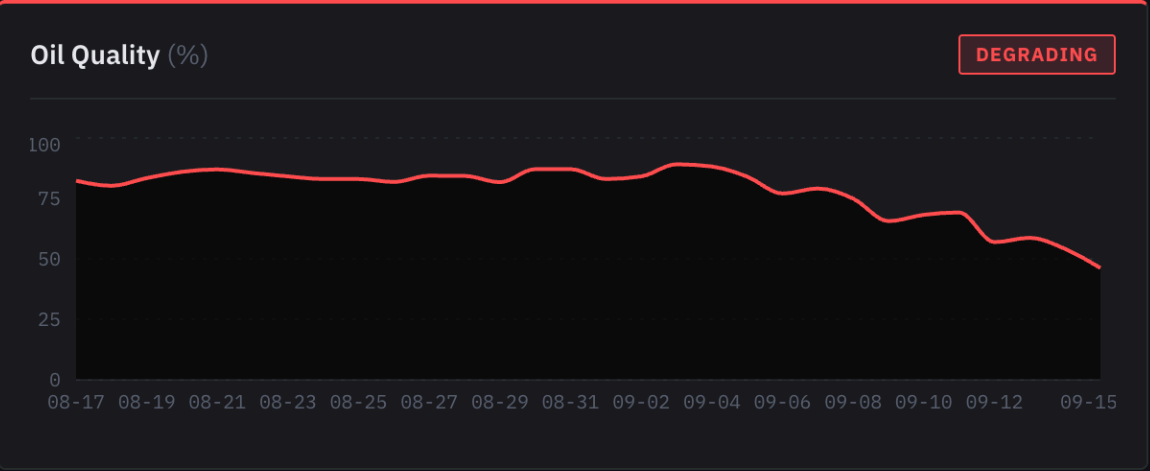
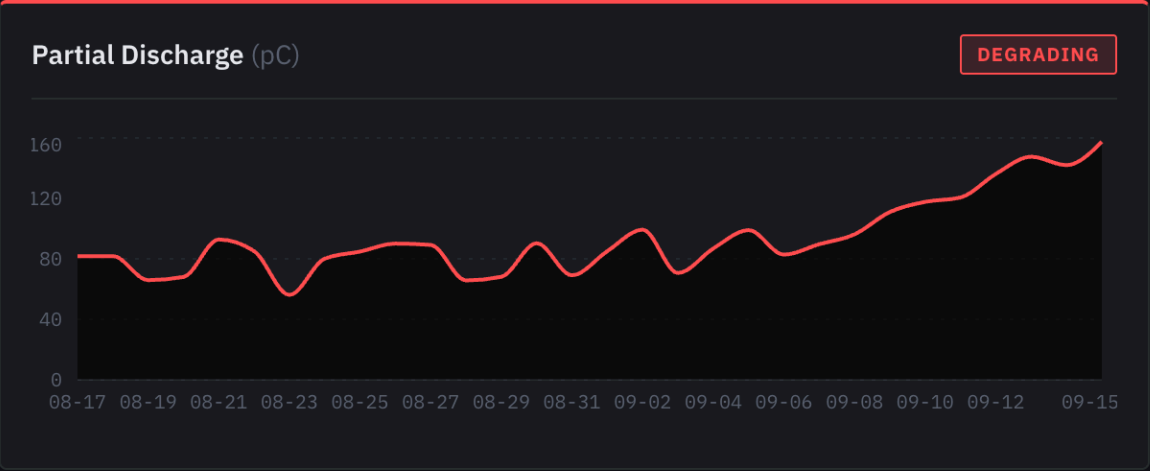
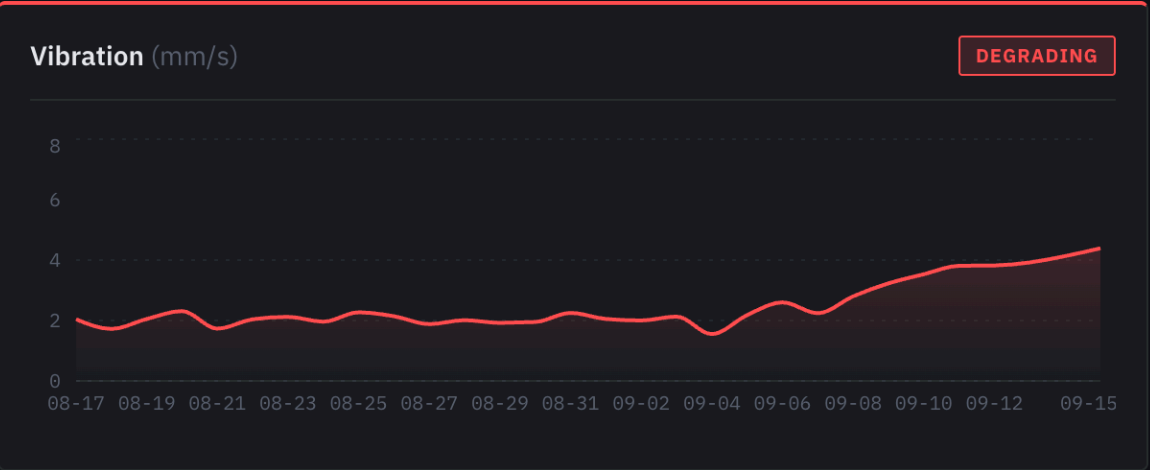
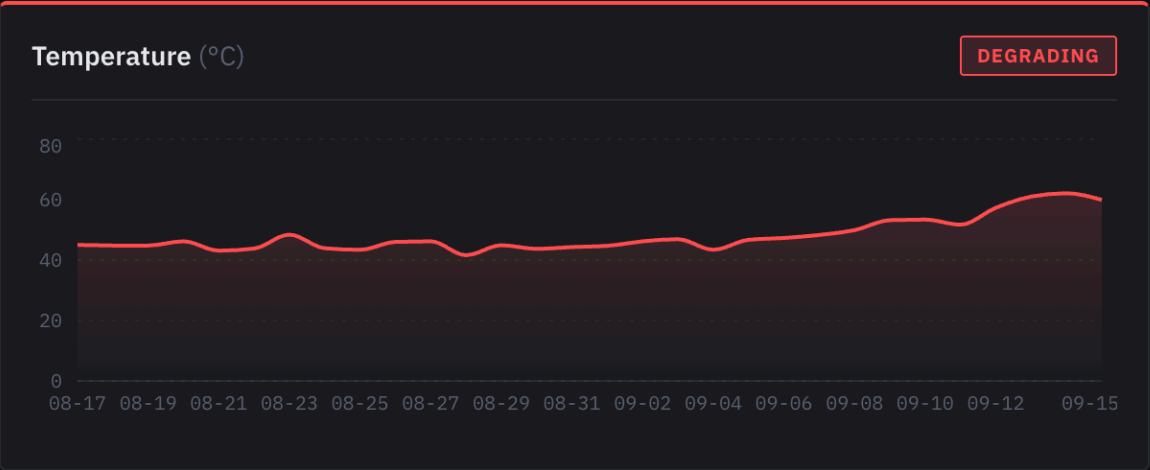
Sensor Anomaly – Temperature

+9 pts

Weather Risk

+7.1 pts

Sensor Trends



7-Day Weather Context — Bayview

09-15 82.3° Wind 14.5 mph Precip 19%	09-16 93.1° Wind 11.4 mph Precip 16%	09-17 74.7° Wind 5.5 mph Precip 10%	09-18 81.8° Wind 6.6 mph Precip 17%	09-19 72.9° Wind 14 mph Precip 17%	09-20 81° Wind 14.9 mph Precip 19%	09-21 81.4° Wind 6.1 mph Precip 23%
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Generated 15/09/2026, 05:30:00

NORTH VALLEY Storm warning on 2026-09-17 (wind 63.9 mph, 93% precip probability).		
North Valley Substation 11	● CRITICAL score 100	By 2026-09-16
Dispatch crew for emergency inspection now; pre-stage replacement parts/transformer.		1D LEFT
Risk score 100.0 (Critical) x grid impact severity 8/10 (48,708 customers served); pre-positioned ahead of the 2026-09-17 storm warning in North Valley.		
North Valley Transformer 04	● CRITICAL score 92.6	By 2026-09-16
Dispatch crew for emergency inspection now; pre-stage replacement parts/transformer.		1D LEFT
Risk score 92.6 (Critical) x grid impact severity 7/10 (6,477 customers served); pre-positioned ahead of the 2026-09-17 storm warning in North Valley.		
North Valley Substation 13	● HIGH score 50.2	By 2026-09-16
Schedule priority inspection within days; order replacement parts if degradation confirmed.		1D LEFT
Risk score 50.2 (High) x grid impact severity 8/10 (57,855 customers served); pre-positioned ahead of the 2026-09-17 storm warning in North Valley.		
North Valley Substation 07	● MEDIUM score 48.8	By 2026-09-16
Schedule inspection in the next routine maintenance window; increase monitoring cadence.		1D LEFT
Risk score 48.8 (Medium) x grid impact severity 8/10 (54,935 customers served); pre-positioned ahead of the 2026-09-17 storm warning in North Valley.		
North Valley Transformer 03	● MEDIUM score 40.2	By 2026-09-16
Schedule inspection in the next routine maintenance window; increase monitoring cadence.		1D LEFT
Risk score 40.2 (Medium) x grid impact severity 9/10 (12,043 customers served); pre-positioned ahead of the 2026-09-17 storm warning in North Valley.		

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Highland Transformer 05

TYPE

transformer

REGION

Highland

INSTALLED

2023

CAPACITY

6.8 MVA

CUSTOMERS SERVED

1,843

GRID IMPACT

6/10

Why This Score

Weather Risk

No storm warning for Highland, but elevated wind/precipitation in the 7-day forecast add modest risk.

Baseline Operational Risk

Fixed floor reflecting that any energized grid asset carries nonzero risk.

Historical Incident Rate

Transformers of similar age (3 yrs, 'new' bracket) show 1 historical incidents in the record; base failure rate estimated at 2.0%/yr from age and asset class.

Sensor Anomaly — Temperature

Temperature is stable within its normal operating range.

Sensor Anomaly — Vibration

Vibration is stable within its normal operating range.

Sensor Anomaly — Partial Discharge

Partial Discharge is stable within its normal operating range.

Grid Copilot



highest risk assets

Top 5 highest-risk assets: - Eastgate Transformer 04 — Critical (score 100.0/100, Eastgate) - North Valley Substation 11 — Critical (score 100.0/100, North Valley) - Eastgate Transformer 03 — Critical (score 97.6/100, Eastgate) - North Valley Transformer 04 — Critical (score 92.6/100, North Valley) - Eastgate Transformer 02 — Critical (score 90.5/100, Eastgate)

▼ Tools called (1)

get_at_risk_assets
{ "region": null, "min_tier": "High", "limit": 10 }

build a maintenance plan

Top prioritized maintenance actions: - North Valley Substation 11 (North Valley) — Dispatch crew for emergency inspection now; pre-stage replacement parts/transformer. by 2026-09-16 - Eastgate Transformer 04 (Eastgate) — Dispatch crew for emergency inspection now; pre-stage replacement parts/transformer. by 2026-09-17 - Eastgate Transformer 03 (Eastgate) — Dispatch crew for emergency inspection now; pre-stage replacement parts/transformer. by 2026-09-17 - Highland Substation 03 (Highland) — Dispatch crew for emergency inspection now; pre-stage replacement parts/transformer. by 2026-09-16 - North Valley Transformer 04 (North Valley) — Dispatch crew for emergency inspection now; pre-stage replacement parts/transformer. by 2026-09-16

Ask the grid copilot...

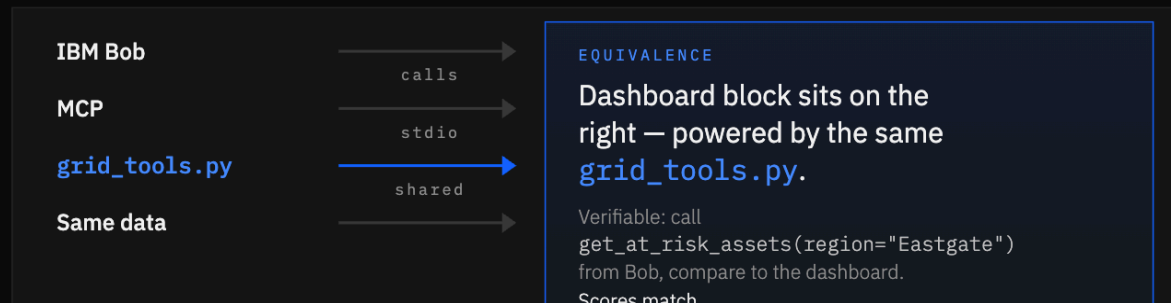
Send

IBM Bob integration is load-bearing — not a README mention.

COLUMN A

IBM Bob / MCP

- Runs as a standalone MCP server (`python -m app.mcp.server`)
- Exposes 5 tools over stdio
 - `get_at_risk_assets`
 - `get_asset_detail`
 - `explain_asset_risk`
 - `get_maintenance_plan`
 - `list_regions`
- IBM Bob (or any MCP client) calls these and gets the **exact same numbers** as the dashboard — because they call `grid_tools.py`, not a separate implementation.



COLUMN B

Design System

TYPOGRAPHY

IBM Plex Sans + IBM Plex Mono

Used throughout the UI for both reading rhythm and tablet-style data labels.

LAYOUT LANGUAGE

IBM Carbon · Dark mode

IBM Blue `#0f62fe` accent, sharp 2px corners, data-dense layout — built for operators reading on shift.

EXPLAINABILITY

Explainable risk design

Every risk score is broken into named, traceable components with point values — operators can defend every dispatch.

FASTMCP

FastMCP — Python framework for building MCP servers.

Wraps our existing tool functions — no re-implementation, no separate "demo mode."

The right asset, the right action — before the storm hits.



METRIC 01

50

Assets Monitored

Transformers + substations across multiple regions, ranked in real time.



METRIC 02

4

Data Signals Fused

Sensor trends + weather + incident history + grid-impact severity.



METRIC 03

0

Black-Box Scores

Every score broken into named components with point values.



METRIC 04

1

Shared Truth

MCP server & web Copilot call identical code — Bob's answer = dashboard.

AVOIDED OUTCOME

A transformer with a degrading oil-quality sensor and a storm in its region's 7-day forecast is flagged as **Critical** and gets a crew pre-positioned the day before the storm — not an emergency dispatch call during it.

BUSINESS IMPACT

- Preventable outage cost: **\$1M+/hour avoided**
- Misallocated maintenance spend reduced — inspect what needs it, not by calendar.
- Hospitals and critical-load customers **protected with advance warning**

Built in 11 hours at BOB Hackathon

TEAM NAME

MEMBERS

Fantastic Four

04

01 · BACKEND

SR

Sunny Radadiya

Backend Architecture & Risk Engine

- Designed and built `risk_engine.py` (explainable composite scoring)
- Built `maintenance_planner.py` (weather-timed prioritization)
- Set up FastAPI service, Pydantic schemas, and CORS

02 · IBM BOB / MCP

JS

Jenil Sutariya

IBM Bob / MCP Integration & Copilot

- Implemented `app/mcp/server.py` using FastMCP
- Built `copilot_service.py` (deterministic intent router)
- Designed `grid_tools.py` — the shared single source of truth

03 · FRONTEND

KV

Kaushal Vora

Frontend — Dashboard & Asset Detail

- Built React dashboard with KPI chips, risk table, live filters
- Asset Detail page: score focal point, contribution bars, sensor charts, weather cards
- Integrated Recharts for sensor trend visualization

04 · FRONTEND & DATA

VB

Veer Bhalodia

Frontend — Maintenance Plan, UI Design & Data

- Maintenance Plan page with urgency badges and weather-timed deadlines
- Designed dark-mode UI system (tokens, elevation layers, animations)
- Built Grid Copilot panel; authored `generate_synthetic_data.py`

REPOSITORY

github.com/Sunny2307/bob-ai-hackathon-Fantsactic_Four.git